

# Homework #2 - Spaceflight Mechanics

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## 1 Part 1

### 1.1 Problem

On the 31/01/2008 at 10:00:00 (UT) an artificial satellite is located on the Cape Kennedy vertical (28.5° N, 80.6° W) in a circular orbit with radius 6600 km: the azimuth is 90 ° (as it's pointing East). Design the maneuver needed to place the spacecraft in a geostationary orbit on the meridian passing through Cape Kennedy within 2 days from the departure time, using the most efficient strategy. Proceed in steps. Define a mission outline, with one plane change and neglecting the final positioning of the spacecraft on Cape Kennedy meridian. Find the position and time of the significant points along the flight path as well as the parameters of the different parts of the trajectory.

### 1.2 Preliminary data

- LEO orbit

$$r_{LEO} = 6600 \text{ km}$$

$$v_{LEO} = \sqrt{\frac{\mu}{r_{LEO}}}$$

$$T_{LEO} = 2\pi\sqrt{\frac{r_{LEO}^3}{\mu}}$$

- GEO orbit

$$r_{GEO} = \sqrt[3]{\mu \left(\frac{T_{sid}}{2\pi}\right)^2}$$

$$v_{GEO} = \sqrt{\frac{\mu}{r_{GEO}}}$$

$$T_{GEO} = T_{sid}$$

- GTO transfer

$$a_{GTO} = \frac{r_{GEO} + r_{LEO}}{2}$$

$$e_{GTO} = \frac{r_{GEO} - r_{LEO}}{r_{GEO} + r_{LEO}}$$

$$v_{GTO,per} = \sqrt{\frac{\mu}{a_{GTO}}} \cdot \sqrt{\frac{1 + e_{GTO}}{1 - e_{GTO}}}$$

$$v_{GTO,apo} = \sqrt{\frac{\mu}{a_{GTO}}} \cdot \sqrt{\frac{1 - e_{GTO}}{1 + e_{GTO}}}$$

$$T_{GTO} = 2\pi\sqrt{\frac{a_{GTO}^3}{\mu}}$$

|     | $a$      | $e$    | $i$      | $r_{per}$ | $r_{apo}$ | $v_{per}$    | $v_{apo}$   | $T$     |
|-----|----------|--------|----------|-----------|-----------|--------------|-------------|---------|
| LEO | 6600 km  | 0      | 28.5 deg | 6600 km   | 6600 km   | 7.7714 km/s  | 7.7714 km/s | 5336 s  |
| GEO | 42164 km | 0      | 0 deg    | 42164 km  | 42164 km  | 3.0747 km/s  | 3.0747 km/s | 86164 s |
| GTO | 24382 km | 0.7293 | 28.5 deg | 6600 km   | 42164 km  | 10.2196 km/s | 1.5997 km/s | 37889 s |

TABLE 1. Mission outline data

### 1.3 Time calculation

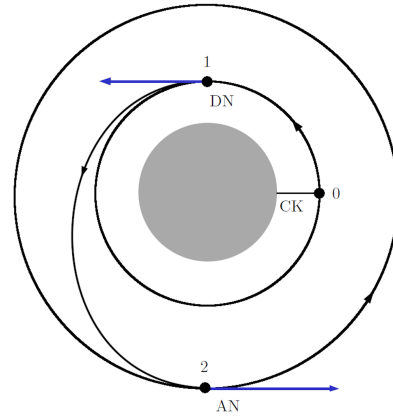
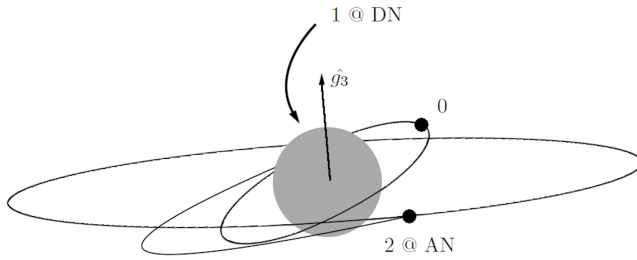
We first need to convert the time given to MJD as to simplify further calculations.

$$\begin{aligned}
 01/01/2008 \text{ @ } 00:00:00 &\Rightarrow MJD = 54466 \\
 31/01/2008 \text{ @ } 00:00:00 &\Rightarrow MJD = 54496
 \end{aligned}$$

We start counting time from the initial position of the spacecraft, so  $t_0$  is zero,  $t_1$  is the latter plus a quarter of the LEO orbit and so on... To do this we added 10 hours to the last measurement of time and shifted the origin of the time axis to match  $t_0$ .

$$\begin{aligned}
 t_0 &= 0 \text{ hrs} & 31/01/2008 \text{ @ } 10:00:00 \\
 t_1 &= t_0 + \frac{T_{LEO}}{4} = 0.3706 \text{ hrs} & 31/01/2008 \text{ @ } 10:22:14 \\
 t_2 &= t_1 + \frac{T_{GEO}}{2} = 5.6330 \text{ hrs} & 31/01/2008 \text{ @ } 15:37:50
 \end{aligned}$$

$$\Rightarrow TOF = t_2 - t_0 = 5.6330 \text{ hrs}$$



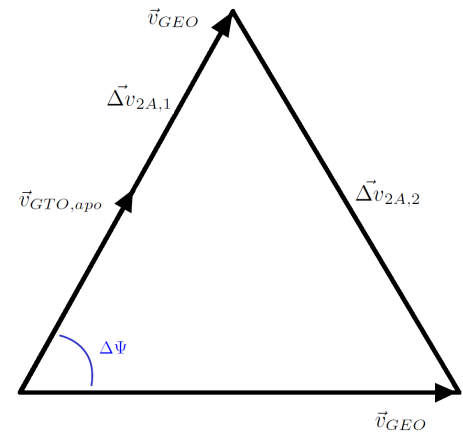
### 1.4 Three different strategies

- Strategy A  
 $\frac{1}{4}$  of LEO orbit until DN, beginning of the Hohmann transfer at 1, end of the Hohmann transfer plus rotation of plane at 2 (in this precise order).

$$\Delta V_{1A} = v_{GTO,per} - v_{LEO}$$

$$\Delta V_{2A} = (v_{GEO} - v_{GTO,apo}) + (2 v_{GEO} \sin\left(\frac{\Delta\Psi}{2}\right))$$

$$\Delta V_A = \Delta V_{1A} + \Delta V_{2A} = 5.4369 \text{ km/s}$$

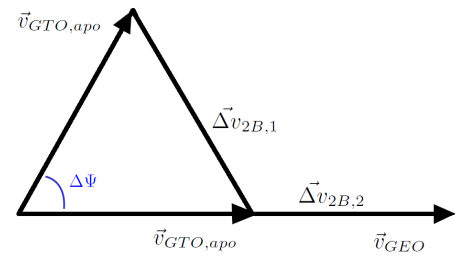


- Strategy B  
 $\frac{1}{4}$  of LEO orbit until DN, beginning of the Hohmann transfer at 1, rotation of plane plus end of the Hohmann transfer at 2 (in this precise order).

$$\Delta V_{1B} = v_{GTO,per} - v_{LEO}$$

$$\Delta V_{2B} = (2 v_{GEO} \sin\left(\frac{\Delta\Psi}{2}\right)) + (v_{GEO} - v_{GTO,apo})$$

$$\Delta V_B = \Delta V_{1B} + \Delta V_{2B} = 4.7107 \text{ km/s}$$

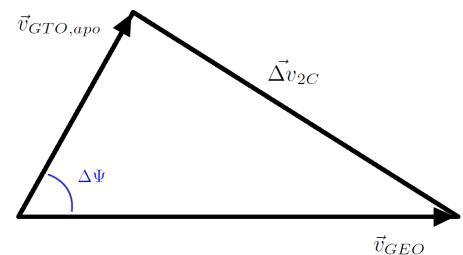


- Strategy C  
 $\frac{1}{4}$  of LEO orbit until DN, beginning of the Hohmann transfer at 1 and combined maneuver (both end of the Hohmann transfer and rotation of plane) at 2.

$$\Delta V_{1C} = v_{GTO,per} - v_{LEO}$$

$$\Delta V_{2C} = \sqrt{v_{GTO,apo}^2 + v_{GEO}^2 - 2 v_{GTO,apo} v_{GEO} \cos(\Delta\Psi)}$$

$$\Delta V_C = \Delta V_{1C} + \Delta V_{2C} = 4.2833 \text{ km/s}$$



## 2 Part 2

### 2.1 Problem

Find the inclination of the GTO (Geostationary transfer orbit) that minimizes  $\Delta V_{tot}$ .

### 2.2 Minization of $\Delta V$

- Strategy D

Combining both maneuvers, the efficiency of the total transfer increases. But, for this to occur we need to divide the rotation in two parts and find the values that make the total  $\Delta V$  minimum.  $28.5^\circ$  is the 10% of  $\Delta\Psi$ , we'll use this value as upper limit to the division.

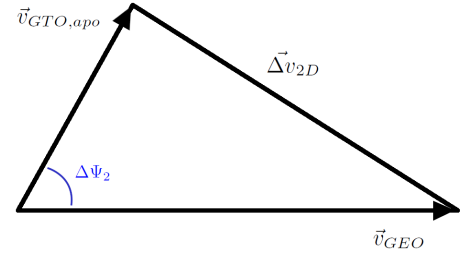
$$\Delta\Psi_1 = \left[0, \dots, 28.5^\circ \cdot \frac{\pi}{180^\circ}\right]$$

$$\Delta\Psi_2 = 28.5^\circ \cdot \frac{\pi}{180^\circ} - \Delta\Psi_1$$

$$\Delta V_{1D} = \sqrt{v_{LEO}^2 + v_{GTO,per}^2 - 2 v_{LEO} v_{GTO,per} \cos(\Delta\Psi_1)}$$

$$\Delta V_{2D} = \sqrt{v_{GTO,apo}^2 + v_{GEO}^2 - 2 v_{GTO,apo} v_{GEO} \cos(\Delta\Psi_2)}$$

$$\Delta V_D = \Delta V_{1D} + \Delta V_{2D}$$



Looking the following graph we are able to understand what values of  $\Delta\Psi_1$  make  $\Delta V_D$  minimum.

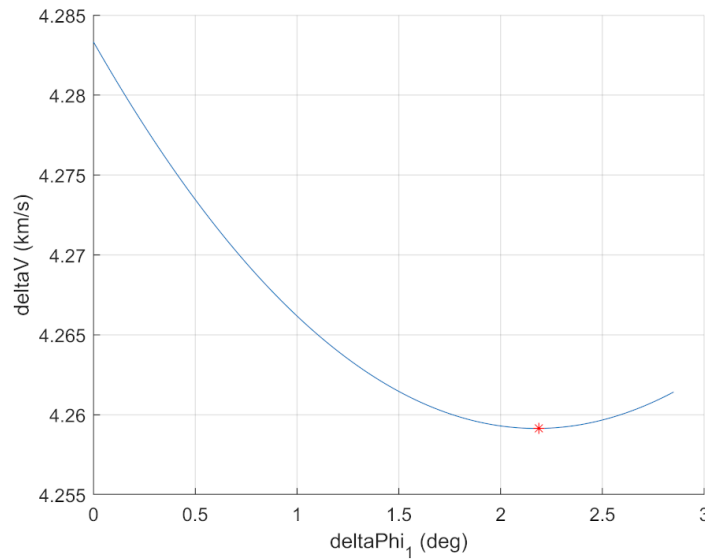


FIGURE 1.  $\Delta\Psi_1$  vs  $\Delta V_D$

$$\Delta\Psi_1 = 2.1879^\circ$$

$$\Delta\Psi_2 = 26.3121^\circ$$

$$\Delta V_D = 4.2591 \text{ km/s}$$

### 3 Part 3

#### 3.1 Problem

Split one of the two  $\Delta V$  in such a way that the request on the final positioning of the spacecraft is fulfilled.

#### 3.2 More strategies

We need to design another orbit capable of lasting the remaining time to ensure the spacecraft enters into final orbit when the meridian passing through Cape Kennedy is beneath it. We're gonna call it *waiting orbit*, or WO.

$$\begin{aligned}
 T_{WO} &= \frac{3}{4}T_{sid} - TOF = 44344 \text{ s} \\
 a_{WO} &= \sqrt[3]{\mu \left( \frac{T_{WO}}{2\pi} \right)^2} = 27078 \text{ km} \\
 v_{WO,apo} &= \sqrt{\frac{2\mu}{r_{GEO}} - \frac{\mu}{a_{WO}}} = 2.0461 \text{ km/s} \\
 e_{WO} &= \frac{r_{GEO}}{a_{WO}} - 1 = 0.5571 \\
 r_{WO,apo} &= a_{WO}(1 - e_{WO}) = 11992 \text{ km}
 \end{aligned}$$

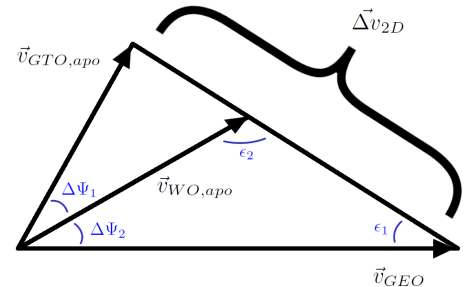
- Strategy E

It's a modification of the strategy C. The spacecraft is put into the waiting orbit, which is used also to rotate the plane twice: one at the the beginning of the waiting orbit and one at its end.

$$\begin{aligned}
 \frac{v_{GTO,apo}}{\sin(\epsilon_1)} &= \frac{\Delta V_{2C}}{\sin(\Delta\Psi)} \quad \Rightarrow \quad \epsilon_1 = \arcsin \left( \frac{v_{GTO,apo}}{\Delta V_{2C}} \cdot \sin(\Delta\Psi) \right) \\
 \frac{v_{GEO}}{\sin(\epsilon_2)} &= \frac{v_{WO,apo}}{\sin(\epsilon_2)} \quad \Rightarrow \quad \epsilon_2 = \pi - \arcsin \left( \frac{v_{GEO}}{v_{WO,apo}} \cdot \sin(\epsilon_1) \right)
 \end{aligned}$$

$$\begin{aligned}
 \Delta\Psi_2 &= \pi - \epsilon_1 - \epsilon_2 = 14.3947^\circ \\
 \Delta\Psi_1 &= \Delta\Psi - \Delta\Psi_1 = 14.1053^\circ
 \end{aligned}$$

$$\begin{aligned}
 \Delta V_{1E} &= v_{GTO,per} - v_{LEO} \\
 \Delta V_{2E} &= \sqrt{v_{GTO,apo}^2 + v_{WO,apo}^2 - 2 v_{GTO,apo} v_{WO,apo} \cos(\Delta\Psi_1)} \\
 \Delta V_{3E} &= \sqrt{v_{WO,apo}^2 + v_{GEO}^2 - 2 v_{WO,apo} v_{GEO} \cos(\Delta\Psi_2)} \\
 \Delta V_E &= \Delta V_{1E} + \Delta V_{2E} + \Delta V_{3E} = 4.2833 \text{ km/s}
 \end{aligned}$$



- Strategy F

3 rotations instead of just 2. The first rotation is the first one of the strategy D, while the remaining one is divided in two, one to be applied at the beginning of the waiting orbit and one at its end.

$$\Delta\Psi_1 = \Delta\Psi_{1D} = 2.1879^\circ$$

$$\epsilon_3 = \arcsin\left(\frac{v_{GTO,apo}}{\Delta V_{2D}} \cdot \sin(\Delta\Psi_{2D})\right)$$

$$\epsilon_4 = \pi - \arcsin\left(\frac{v_{GEO}}{v_{WO,apo}} \cdot \sin(\epsilon_3)\right)$$

$$\Delta\Psi_3 = \pi - \epsilon_3 - \epsilon_4 = 13.0920^\circ$$

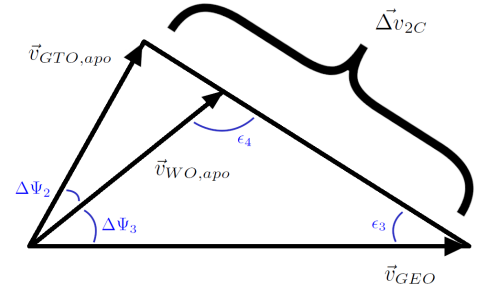
$$\Delta\Psi_2 = \Delta\Psi_{2D} - \Delta\Psi_3 = 13.2202^\circ$$

$$\Delta V_{1F} = \Delta V_{1D}$$

$$\Delta V_{2F} = \sqrt{v_{GTO,apo}^2 + v_{WO,apo}^2 - 2 v_{GTO,apo} v_{WO,apo} \cos(\Delta\Psi_2)}$$

$$\Delta V_{3F} = \sqrt{v_{WO,apo}^2 + v_{GEO}^2 - 2 v_{WO,apo} v_{GEO} \cos(\Delta\Psi_3)}$$

$$\Delta V_F = \Delta V_{1F} + \Delta V_{2F} + \Delta V_{3F} = 4.2591 \text{ km/s}$$



Even if both strategies D and F yields results values very similar, the one provided by strategy F is smaller, and thus more efficient.